

Power Supply Architectures for Drones - A Review

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Abstract—Drones are recently receiving a growing attention in both civil and military sectors. Despite their good features such as high maneuverability, wide variety of usage, and low cost; battery-powered drones are still limited in terms of endurance. They cannot perform long flights and persistent missions. This paper proposes then a review-based discussion of the solutions addressing this issue, including swapping laser-beam inflight recharging and tethering.

Hybrid power supply system is also a solution of choice. Combining battery with different sources such as fuel cell, solar cells, and supercapacitor allows the system to benefit from sources advantages and cover their limitations. In this context, this paper provides a comparative and critical study of different power supply architectures, thus facilitating the trade-off in the choice of the suitable drone power supply system. Insights and recommendations for future research are also provided.

Index Terms—Drone, power supply, swapping, laser-beam inflight recharging, tethered drone.

I. INTRODUCTION

In recent years, drones have received great interest by the scientific community as flying robots with high mobility and low cost [1]. Advancements in drone technology make them smart tools used in several applications [2]: delivery, monitoring, border control, wireless covering, and precision agriculture. In 2020, the drone market value is estimated to be over US\$127 billions [3].

Electric propulsion is favored for drones due to many advantages such as efficiency, reliability, reduced noise and thermal signatures and precise control [4]. The internal combustion engine (ICE) unfortunately misses these key features, and that prevents it from being the solution of choice despite its very high power and energy densities. Hybridization of an ICE with an electric motor enables the drone to benefit from both engines advantages [5], [6]. However, decreasing pollutants and greenhouse gases emissions in the aviation field becomes a high-priority criteria.

Many commercial drones are powered by batteries, their flight time reaches up to two hours. However, despite all advancements in batteries technologies, they are still unable to ensure long and persistent drones missions when used as unique energy source. It is mainly due to their relatively low energy density [7]. Multiplying batteries is unfortunately not a practical solution, their number is limited by weight and space constraints. Thus, drones have to carry an additional energy source which can cover batteries limitations and extend the

endurance. In this context, a fuel cell is typically integrated due its features such as high energy density and fast refueling. In majority of hybrid electrical drones, the fuel cell is selected as main power source, when the other sources are auxiliaries supplying in special flight conditions [8]–[10]. Solar energy is also a good option for fixed-wing drones since they are able to carry solar cells. Therefore, the flight time can be extremely extended to reach many days, and hydrogen can be saved [11]–[15]. During the flight, drones are supposed to perform different maneuvers such as take-off and climbing, then peak power has to be supplied instantaneously. In this context, a supercapacitor can take part in the power supply system thanks to its very high specific power [16], [17]. Consequently, hybridization is a key option for drones to have good performances when ensuring a large operating time. The choice of power supply system components is then important, both of power sources characteristics and mission type have to be considered. In addition, implementation of an energy management system (EMS) is necessary to keep all sources operating with efficiency and to preserve their lifetimes.

Other approaches are available for battery-based drones allowing them to operate continuously: swapping, laser-beam inflight recharging, and tethering [18]–[24]. Swapping offers the possibility to recharge the drone's depleted batteries during its mission using docking stations. Light energy can be transferred to drones in-flight using laser beam, batteries can then be charged after converting light power to electricity. Tethered drones are continuously supplied through power lines, thereby they can have unlimited endurance.

In this paper, the attention is focused on the drone power supply. We propose a comprehensive and critical evaluation of drones available power supply structures and techniques providing insights about the choice of the power supply system. At the end of the paper, some useful guiding recommendations and prospects will be proposed.

Section 2 discusses battery-based drones supplying techniques. Section 3 investigates the fuel cell as main power source. Section 4 presents and evaluates different possible hybrid structures used in drones power supply systems.

II. BATTERY-POWERED DRONES

Batteries constitute the pillar of battery-powered drones. They supply most small drones offering flexibility and simplicity for the propulsion system. However, LiPo batteries

for example, can ensure drone operation for a maximum of 90 min [8]. Such limitation restricts these drones usage to commercial purposes. Donato et al. [4] proposed different battery technologies evaluation considering the state of charge (SOC) for a given mission. In [25], battery-powered drones performances were investigated and mathematical formulation for range and endurance estimation was proposed.

One of the most important battery-powered drones challenges is their reduced autonomy. Even with current advancements in batteries technologies, flight endurance and range are still limited by the battery specific energy. This section will address promising techniques used to allow the battery-based drones to recharge or replace their depleted batteries during the flight, and then extremely increase endurance.

A. Battery-based Supplying Techniques

1) *Swapping*: Drones can recharge or replace their depleted batteries during their mission using swapping technique. This task can be autonomously conducted or human-operated. By using drone swarm and by managing their cooperation, the swapping-based multi-agent system can operate continuously performing persistent missions. A typical swapping operation need a ground recharge station for batteries charging/replacing. It can be deployed on cell towers, rooftops, power poles, or standalone pylons [26], [27]. This docking platform can be fed by big batteries, power lines, or solar cells for remote operation. Figure 1 illustrates the swapping and hotswapping approaches and Fig. 2 shows some commercially available ground stations (GSs).

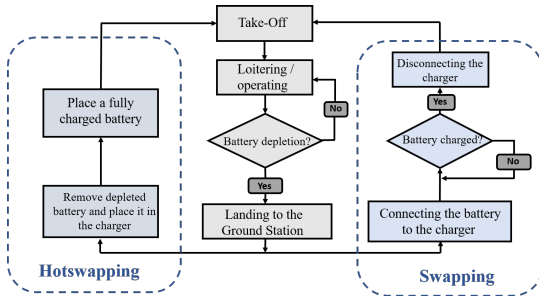


Fig. 1. Swapping and hotswapping algorithms.

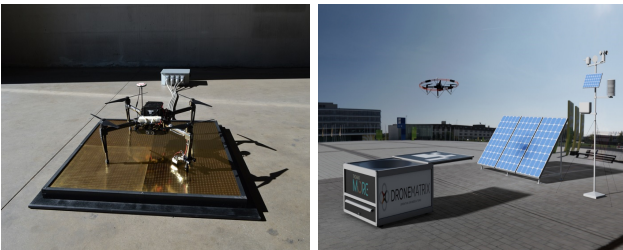


Fig. 2. Drones docking stations [28], [29].

2) *Laser-beam inflight recharging*: Beside the swapping approach, wireless recharging is also proposed as solution. Drones can be charged while flying, without the need of

landing maneuvers, thus increasing the operation safety and efficiency [22], [30]. A prime power source should be installed in the GS to supply the laser generator which transmits the laser-beam to the drone's optical receiver. When the battery becomes discharged, an aerial power link will be established between drone and the nearest laser source (Fig. 3). To avoid laser-beam obstruction, GSs are deployed on rooftops of high buildings or on mobile stations. LaserMotive developed a working prototype of hundreds of watts [20]. In addition, more than 12 h flight was conducted by a quadcopter in [21] proving the feasibility of the proposed approach. This study considered size, payload, and specific drone application. The flight control system and the mechanical design were also presented.

Using this technique, the drone has to operate in a limited area in order to keep the radiative link with the laser generator. It is worth noting that the drone's operating height is legally limited by 120 m according to Federal Aviation Administration (FAA) [31]. Thus, the operating range will be also limited by this constraint. Moreover, each drone need its dedicated laser source, therefore it is necessary to decrease drones number to have a reasonable operational cost [22].

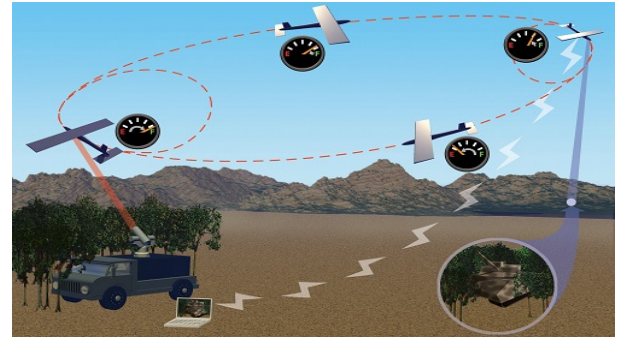


Fig. 3. Laser-beam powered drones [32].

3) *Tethered drones*: Drones can be attached to a ground power source using connection lines to have unlimited autonomy. This option ensures an uninterrupted electricity supply and prevents the drone operation to be disturbed by repetitive recharging and extremely reducing batteries weight. It enables also real time data transfer with safety and rapidity, especially when using fiber technology. Fiber optic cables allow a power transfer of many kilowatts using high-intensity light and decreasing the detectability. Figure 4 provides an illustrative example of tethered drone system for communication and surveillance missions.

In [23], a tethered drone is used on the ship to detect maritime pollution caused by the oil spilled on the sea. Gu et al. [34] demonstrated prototypes destined to nuclear power plants long missions attaining duration of few months. However, a tethered drone can operate only in a limited area since it has to be attached to the ground station. A moving prime power source can be a solution to this issue allowing the drone to cover a larger area.

Table I proposes a critical comparison of the battery-based power supplying techniques.

TABLE I
COMPARISON BETWEEN DRONES BATTERY-BASED SUPPLYING TECHNIQUES.

Power supplying technique	Advantages	Limitations and drawbacks	Related papers
Swapping	Continuous operation, appropriate to long-range flights, battery is the unique power source $\Rightarrow$ Weight can be greatly reduced and power management system is not needed.	The ground station (GS) is mandatory, operational efficiency is affected by the necessity of landing during mission, multiplying drones and batteries $\Rightarrow$ High operational cost, complexity of cooperation between drones and the GS.	[18], [19], [35]
Inflight charging	Continuous operation, wireless refueling, landing during flight is avoided, unique power source, suitable to persistent missions.	Necessity of GS, reduced operating range and height, concerns of laser-beam obstruction.	[20]–[22], [30]
Tethered drones	Continuous operation, docking is not required, one energy source, real-time data transfer with safety and efficiency, persistent operation.	The GS is required, operating area is limited, risk of drone damage when tethering is lost.	[23], [24]



Fig. 4. Tethered drone for communication and surveillance [33].

III. FUEL CELL POWERED DRONES

Fuel cells when supplying drones, as illustrated by Fig. 5, can extend the endurance to hours thanks to their higher specific energy [36]. In addition, they are refueled almost instantly, while batteries recharging process takes a longer time. Using Protonex 550 W PEM fuel cell, the developed drone in [37] showed in a flight test 6 times endurance increase than Li-Ion battery system. Indeed, fuel cells are characterized by a lower energy density, since the fuel tank volume has to be considered.



Fig. 5. Fuel cell based drone from Intelligent Energy [38].

A. Fuel Cells Efficiency Issue

Elements on the process of electricity production in fuel cells can be found in [36], [39]. Fuel cells can reach a maximum efficiency level of 60% [35]. It is still lower than that of lithium batteries (over 90%). That is mainly due to the auxiliary subsystems needed for fuel cell stack operation as illustrated by Fig. 6 [36].

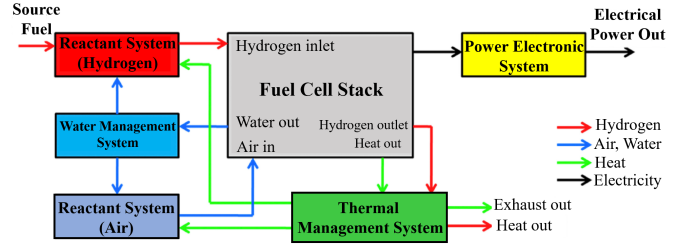


Fig. 6. Fuel cell system subsystems illustration [40].

B. Fuel Storage

Hydrogen has a density of only 0.089 kg/m^3 at standard temperature and pressure [7], and it is not possible to store hydrogen under extremely high pressure and low temperature [41]. Three techniques are mainly used to overcome this constraint: compressed hydrogen gas, liquid hydrogen, and chemical hydrogen generation.

In [37], Swider-Lyons et al. proposed a comparative study of hydrogen storage methods in the Ion Tiger drone when conducting a 24 hours flight. According to several research teams, for example, Naval Research Laboratory [42], liquid H_2 can't be the best choice since it requires specific infrastructures [7]. In Colorado State University [10], compressed H_2 based drone succeed to complete 24 hours flight-test. As regards chemical hydrogen generation, extra equipment are required to extract hydrogen, which make the power system heavy and complex. In addition, hydrogen extraction process makes time, therefore increasing drone response-time to load changes.

IV. HYBRID POWER SOURCES

1) *Fuel cell and battery*: Drones when powered by fuel cells as a unique power source present some limitations.

The process of electricity generation which need air supply using compressors, pumps, and valves increases the time-constant (in the order of seconds) relatively to batteries. Thus, hybridization of fuel cell and battery can lead to a power system with effective performances combining the advantages of both sources and balancing their drawbacks [4], [44]. Indeed, battery will supply the peak power needed in take-off and climbing since it has higher power density and faster response. In cruise or descend periods, fuel cell will supply the main demand power, and it can also charge the battery to keep the SOC in a favorable range.

In [45], authors proposed a hardware-in-the-loop (HIL) simulations of a hybrid drone fed by a 200 W fuel cell and a battery. This study provides an analysis of the behavior of each source using several tests and considering endurance and hydrogen use. In a similar study in [46], load fluctuations were considered in HIL simulations based on real flight data. In their experimental investigation, Gong et al. [47] assessed the battery contribution degree under several solicitations. However, the energy management was not considered in this studies, power was supplied with the passive method.

2) *Solar cells as extra power source*: Application of solar power in drones becomes fundamental for high altitude long endurance (HALE) missions as illustrated by Fig. 7. Photovoltaic (PV) power allows drones when outfitted by battery storage system to achieve multi-day flights carrying a variety of sensors.

In [48], Harvey et al. investigated the effect of adding a PV system to a power supply system based on battery and fuel cell using HIL simulations. It was concluded that using PVs might allow performances improvement and up to 59% of energy saving. As shown by Fig. 7, it is clear that large wings are needed for solar powered-drones to exploit the maximum of available solar energy. In addition, a maximum power point tracking (MPPT) algorithm has to be implemented [49]. In [12], Shiau et al. have experimentally designed a solar power management system (SPMS) for their drone powered by solar cells and batteries. However, this study considers only on-board electronic circuitries as load, the propulsion power was neglected. In [49], Pen et al. have investigated the effect of solar irradiation fluctuations on drones operation. An interesting MPPT algorithm based on perturb and observe (P&O) method was proposed to improve the system efficiency and overcome this climatic constraint.



Fig. 7. Solar-powered drones [50], [51].

3) *Supercapacitor as an extra power source*: Supercapacitors are recently receiving attention in drones power supply system since they present some advantages that can balance batteries limitations [52]. Batteries suffer from lifetime decrease at high temperatures, sluggish charge/discharge, and self-discharging.

A supercapacitor when compared to a battery, has a much higher specific power. In addition, the supercapacitor technology is characterized by a fast charging/discharging speed, overcharge tolerance, and the ability to extremely reduce the DC bus voltage fluctuations [53]. In this context, a supercapacitor will reinforce the hybrid power supply system offering higher power density and allowing rapid power response (Fig. 8). In this case an energy management system (EMS) is necessary to enable optimal operation for each power source.

In [16], Gong et al. evaluated the supercapacitor behavior in a drone with hybrid power supply system including battery and fuel cell. Using HIL simulations, the authors provided a comparison with a fuel cell/battery system considering a flight profile. This study has proved supercapacitor inducing good performances in terms of dynamic response and load smoothing. In another research [17], the authors developed a real drone prototype keeping the same drone configuration. In this study, two dynamic flight tests were conducted to show the supercapacitor significant role in supplying peak power, reducing power fluctuations, and DC bus voltage stabilization. The power splitting method has unfortunately not been discussed.

A critical comparative analysis on drone hybrid power supply configurations is proposed in Table II.

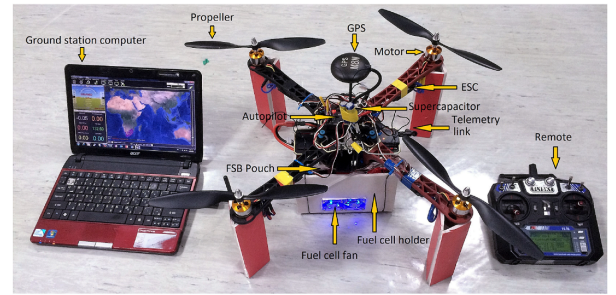


Fig. 8. A hybrid fuel cell-battery-supercapacitor drone system [54].

V. CONCLUSIONS AND FUTURE TRENDS

Drones are receiving big interest from researchers in several domains. As the power supply system is a fundamental unit in a drone platform, this paper has therefore dealt with comparative and critical study on drones supplying sources and architectures. This review-based study aims providing basis and insights about the choice of the appropriate power supply devices considering sources features and flight requirements. It has also discussed supplying solution for specific missions when battery-based drones have to operate continuously, namely swapping, laser-beam inflight recharging, and tethered technologies.

TABLE II
COMPARISON OF HYBRID POWER SUPPLY CONFIGURATIONS.

Sources	Advantages	Limitations	Related papers
Thermal energy (ICE)	Very high specific power and energy, large autonomy and payload range.	Thermal and acoustic signatures, low efficiency, GHG emission, fuel high cost.	[5], [6], [55]
Battery	High specific energy, energy is stored (not generated) $\Rightarrow$ short response time.	Low specific power, reduced flight time, slow recharging, to extend autonomy $\Rightarrow$ multiply batteries $\Rightarrow$ increase in both of weight and cost.	[25], [56]
Fuel cell	High specific power, rapid refueling without "memory effect", to extend endurance $\Rightarrow$ add more fuel to the same stack $\Rightarrow$ weight saved.	Energy is generated $\Rightarrow$ long response time relatively, auxiliary systems are needed (compressors, regulator, etc), safety and hydrogen storage concerns, limited infrastructures for hydrogen supply, costly hydrogen production process.	[9], [10], [37]
Fuel cell and Battery	Energy and power densities are increased $\Rightarrow$ improvements in the endurance and the response time, rapid refueling and energy storage.	Weight and volume increase, EMS is required $\Rightarrow$ additional complexity in control.	[46], [47], [57]–[59]
Fuel cell, Battery, and solar cells	Supplementary energy source $\Rightarrow$ flight time is increased, clean, available, and free power $\Rightarrow$ low energy cost, fuel saving.	Limited to fixed-wing drones, large wings constraint, necessity of an energy storage system, EMS with MPPT algorithm is needed.	[11]–[15]
Fuel cell, Battery, and supercapacitor	Very high power density, fast charging, reduced weight, and reduced dc bus fluctuations; very long lifetime, minimum heat loss due to the reduced internal resistance.	EMS is needed, supercapacitor high cost, supercapacitor voltage regulation is necessary.	[16], [17]

It can be concluded that the choice of power supply components may not be sufficient to have optimal operation with extended endurance. A power management system is then required to optimally split power using optimization strategies while respecting drone specifications. In this context, off-line optimization methods could be proposed as a trend. Some research works are focusing on the energy consumption prediction using a priori flight data [60], [61].

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